CUSTOMER INTELLIGENCE DASHBOARD USING MACHINE LEARNING, FASTAPI AND GENERATIVE AI

Problem Statement

Modern retail businesses generate vast amounts of customer transaction and behavioral data. While traditional analytics systems can provide descriptive statistics and predictive models, organizations often struggle to transform these outputs into actionable business strategies that improve customer engagement, revenue generation, and customer retention.

A retail company seeks to develop an intelligent customer analytics platform capable of understanding customer behaviour, identifying customer segments, predicting future spending patterns, estimating subscription likelihood, and automatically generating business recommendations based on these predictions. The objective is to leverage both Machine Learning and Generative Artificial Intelligence (GenAI) to enable data-driven decision-making and personalized customer engagement strategies.

To address this challenge, an AI-Powered Customer Intelligence Dashboard will be developed using Machine Learning, FastAPI, Generative AI (Google Gemini), and interactive web technologies. The system will analyze customer shopping behaviour, provide real-time predictions, and generate natural-language business insights, marketing recommendations, and customer engagement strategies through an intuitive dashboard.

The project aims to answer the following overarching question:

"How can customer shopping data be leveraged using Machine Learning and Generative AI to automatically segment customers, predict spending behaviour, estimate subscription likelihood, and generate actionable business insights that improve customer engagement, retention, and overall business performance?"

Objectives

1. Identify meaningful customer segments using clustering techniques.
2. Predict customer spending behaviour using supervised learning models.
3. Estimate customer subscription likelihood and retention potential.
4. Generate AI-powered business insights and marketing recommendations using Generative AI.
5. Deliver predictions and insights through a real-time web-based dashboard.
6. Support data-driven business decisions through analytics and visualization.

Expected Outcome

The final system will provide:

- Customer Segmentation using Machine Learning.
- Spending Prediction for individual customers.
- Subscription Likelihood Prediction with probability scores.
- AI-Generated Customer Insights using Google Gemini.
- Automated Business Recommendations.
- Personalized Marketing Strategies.
- Interactive Dashboard for business users.
- Data-driven decision support for customer engagement and revenue optimization.

The solution combines predictive analytics with Generative AI to bridge the gap between data analysis and business action, enabling organizations to make faster, smarter, and more personalized decisions.